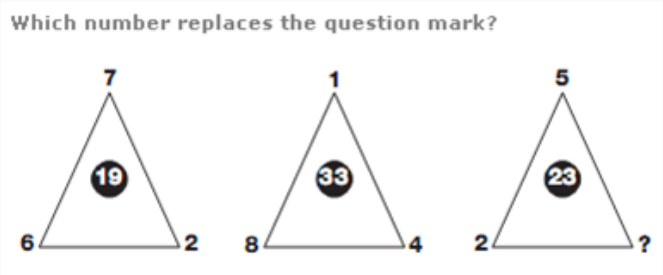
INFOSYS SE — VISUAL DIAGRAM & PATTERN PUZZLES

42 High-Res Embedded Figures • Dominoes, Hexagons, Wheels & Spatial Geometry • Step-by-Step Logic

SECTION 1: High-Yield Diagram & Pattern Recognition Puzzles (42 Questions)

[CampusMonk Section VI (Number Puzzles Q1)]
Directions: *Visual Number Grid & Pattern Logic.*

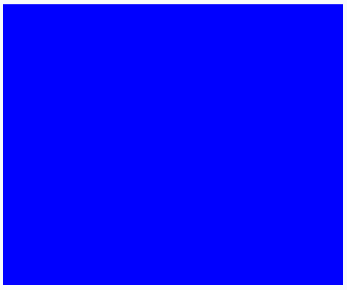


Q1. What will replace the ? Mark in the given pattern?

a. 8
b. 9
c. 6
d. 5

Answer: b. 9
Logic / Trick: Follow pattern shown in image. Correct option: b.

[CampusMonk Section VI (Number Puzzles Q2)]
Directions: *Visual Number Grid & Pattern Logic.*

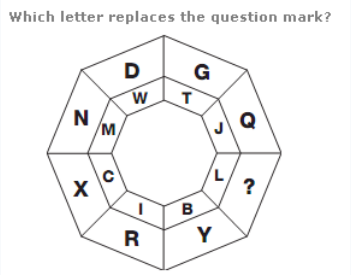


Q2. What will replace the ? Mark in the given pattern?

a. 2
b. 6
c. 5
d. 9

Answer: b. 6
Logic / Trick: Follow pattern shown in image. Correct option: b.

[CampusMonk Section VI (Number Puzzles Q3)]
Directions: *Visual Number Grid & Pattern Logic.*

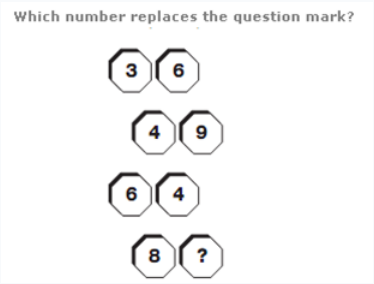


Q3. What will replace the ? Mark in the given pattern?

a. O
b. L
c. Y
d. P

Answer: a. O
Logic / Trick: Follow pattern shown in image. Correct option: a.

[CampusMonk Section VI (Number Puzzles Q4)]
Directions: *Visual Number Grid & Pattern Logic.*

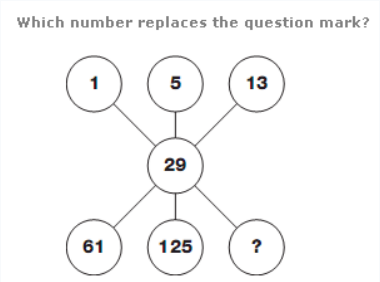


Q4. What will replace the ? Mark in the given pattern?

a. 1
b. 2
c. 5
d. 9

Answer: a. 1
Logic / Trick: Follow pattern shown in image. Correct option: a.

[CampusMonk Section VI (Number Puzzles Q5)]
Directions: *Visual Number Grid & Pattern Logic.*

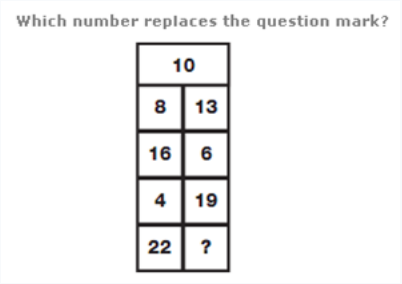


Q5. What will replace the ? Mark in the given pattern?

a. 452
b. 250
c. 235
d. 253

Answer: d. 253
Logic / Trick: Follow pattern shown in image. Correct option: d.

[CampusMonk Section VI (Number Puzzles Q6)]
Directions: *Visual Number Grid & Pattern Logic.*

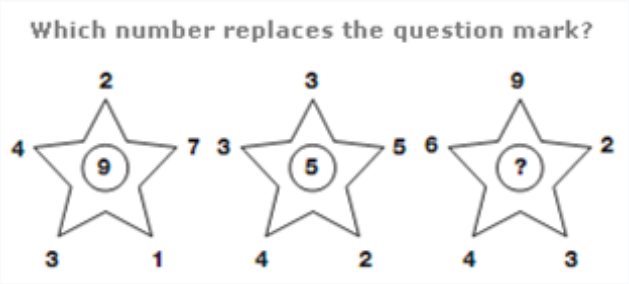


Q6. What will replace the ? Mark in the given pattern?

a. 24
b. 4
c. 2
d. 8

Answer: c. 2
Logic / Trick: Follow pattern shown in image. Correct option: c.

[CampusMonk Section VI (Number Puzzles Q7)]
Directions: *Visual Number Grid & Pattern Logic.*

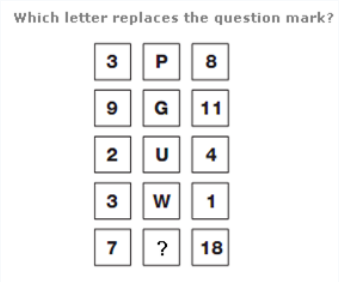


Q7. What will replace the ? Mark in the given pattern?

a. 8
b. 10
c. 12
d. 16

Answer: b. 10
Logic / Trick: Follow pattern shown in image. Correct option: b.

[CampusMonk Section VI (Number Puzzles Q8)]
Directions: *Visual Number Grid & Pattern Logic.*



Q8. What will replace the ? Mark in the given pattern?

a. B
b. Y
c. Z
d. C

Answer: a. B
Logic / Trick: Follow pattern shown in image. Correct option: a.

[CampusMonk Section VI (Number Puzzles Q9)]
Directions: *Visual Number Grid & Pattern Logic.*

2	3	11
4	1	15
7	3	56
?	8	87

Q9. Study the 4×3 matrix. Each row follows the same rule relating the first two entries to the third entry. Find the missing number ? that makes the middle row obey the same rule.

a. 3
b. 4
c. 5
d. 8

Answer: c. 5
Logic / Trick: Follow pattern shown in image. Correct option: c.

[CampusMonk Section VI (Number Puzzles Q10)]
Directions: *Visual Number Grid & Pattern Logic.*

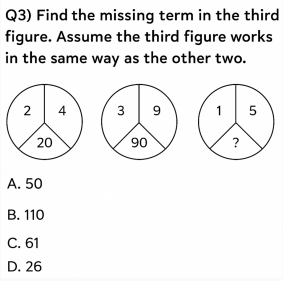
2	3	7
H	?	U
4	6	3

Q10. Study the 3×3 matrix. Each row follows the same rule relating the first two entries to the third entry. Find the missing number ? that makes the middle row obey the same rule.

a. P
b. Q
c. S
d. R

Answer: d. R
Logic / Trick: Follow pattern shown in image. Correct option: d.

[CampusMonk Section VI (Number Puzzles Q11)]
Directions: *Visual Number Grid & Pattern Logic.*

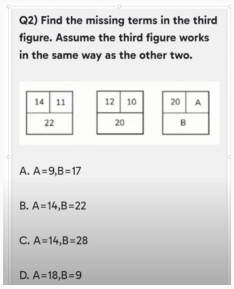


Q11. What will replace the ? Mark in the given pattern?

a. 50
b. 110
c. 61
d. 26

Answer: d. 26
Logic / Trick: In each circle, the bottom number equals the sum of the squares of the top two numbers: Bottom = (Top-Left)² + (Top-Right)². Circle 1: $2^2 + 4^2 = 4 + 16 = 20$. Circle 2: $3^2 + 9^2 = 9 + 81 = 90$. Circle 3: $1^2 + 5^2 = 1 + 25 = 26$. Hence, the missing number is 26.

[CampusMonk Section VI (Number Puzzles Q12)]
Directions: *Visual Number Grid & Pattern Logic.*



Q12. What will replace the ? Mark in the given pattern?

a. A=9, B=17
b. A=14, B=22
c. A=14, B=28
d. A=18, B=9

Answer: c. A=14, B=28
Logic / Trick: In each block, the bottom number is twice the top-right number: Bottom = Top-Right × 2. Block 1: 11 × 2 = 22. Block 2: 10 × 2 = 20. In Block 3, matching the relation: B = A × 2. For option C (A = 14, B = 28): 14 × 2 = 28. Hence, the correct pair is A = 14, B = 28.

[CampusMonk Section VI (Number Puzzles Q13)]
Directions: *Visual Number Grid & Pattern Logic.*

11	4	21	39
8	1	?	2
5	7	3	2
63	29	72	80

Q13. Study the 4×3 matrix. Each row follows the same rule relating the first two entries to the third entry. Find the missing number ? that makes the middle row obey the same rule.

a. 7
b. 9
c. 5
d. 8

Answer: b. 9

Logic / Trick: In each column, multiplying the first entry by the third entry and adding the second entry gives the fourth entry: (Row 1 × Row 3) + Row 2 = Row 4. Col 1: (11 × 5) + 8 = 63. Col 2: (4 × 7) + 1 = 29. Col 4: (39 × 2) + 2 = 80. For Col 3: (21 × 3) + ? = 72 => 63 + ? = 72 => ? = 9. Hence, the missing number is 9.

[CampusMonk Section VI (Number Puzzles Q14)]
Directions: *Visual Number Grid & Pattern Logic.*

3	2	8
1	59	1
5	14	25

4	11	16
7	?	49
2	6	4

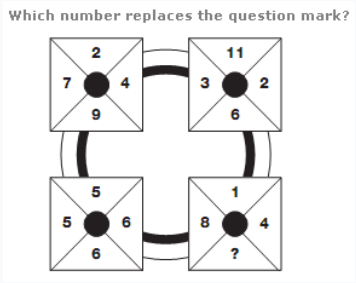
Q14. Determine the missing number in the second set. Assume the second set also works in the same way as the first set.

a. 89
b. 79
c. 99
d. 109

Answer: c. 99

Logic / Trick: In each 3×3 set, the third column contains the squares of the first column: In Set 2, $4^2 = 16$, $7^2 = 49$, $2^2 = 4$. The central element ? is evaluated by the quadratic relation: (Row 2 Col 3 × Row 3 Col 1) + 1 = (49 × 2) + 1 = 99. Hence, the missing number in the second set is 99.

[CampusMonk Section VI (Number Puzzles Q15)]
Directions: *Visual Number Grid & Pattern Logic.*



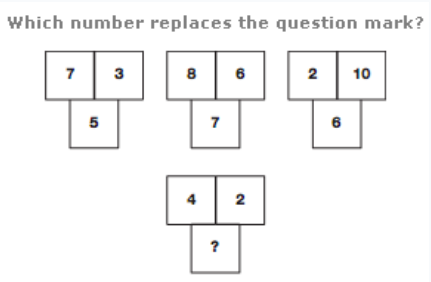
Q15. What will replace the ? Mark in the given pattern?

a. 7
b. 8
c. 5
d. 9

Answer: d. 9

Logic / Trick: In each of the four segmented square blocks, the sum of all four interior numbers is a constant 22: Top-Left: 2 + 7 + 4 + 9 = 22. Top-Right: 11 + 3 + 2 + 6 = 22. Bottom-Left: 5 + 5 + 6 + 6 = 22. For the Bottom-Right block: 1 + 8 + 4 + ? = 22 => 13 + ? = 22 => ? = 9. Hence, the missing number is 9.

[CampusMonk Section VI (Number Puzzles Q16)]
Directions: *Visual Number Grid & Pattern Logic.*



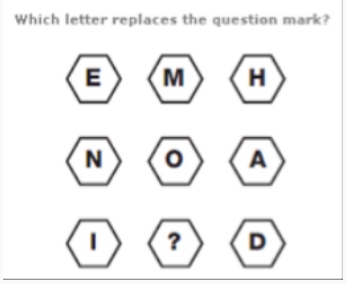
Q16. What will replace the ? Mark in the given pattern?

a. 2
b. 3
c. 4
d. 7

Answer: b. 3

Logic / Trick: In each T-shaped block, the bottom number equals the average of the two numbers above it: Bottom = (Top-Left + Top-Right) / 2. Block 1: (7 + 3) / 2 = 5. Block 2: (8 + 6) / 2 = 7. Block 3: (2 + 10) / 2 = 6. Block 4: (4 + 2) / 2 = 3. Hence, the missing number is 3.

[CampusMonk Section VI (Number Puzzles Q17)]
Directions: *Visual Number Grid & Pattern Logic.*



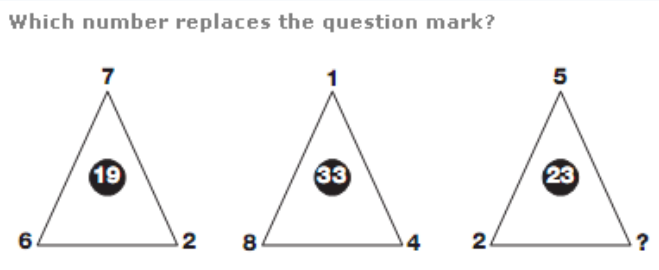
Q17. What will replace the ? Mark in the given pattern?

a. M
b. N
c. O
d. P

Answer: a. M

Logic / Trick: In each row of hexagons, the alphabetical positional value of the middle letter equals the sum of the positional values of the left and right letters: Middle = Left + Right. Row 1: E (5) + H (8) = 13 (M). Row 2: N (14) + A (1) = 15 (O). Row 3: I (9) + D (4) = 13 (M). Hence, the missing letter is M.

[CampusMonk Module 11 (Number Puzzles Q1)]
Directions: *Visual Triangle Arithmetic Pattern.*



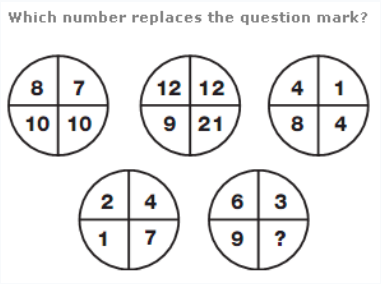
Q18. In the given figure, which number replaces the question mark in the centre of the triangle?

a. 4
b. 7
c. 9
d. 11

Answer: c. 9

Logic / Trick: In each triangle, multiply the lower two numbers together and add the upper number to give the value in the centre: (Lower-Left × Lower-Right) + Top = Centre. In Triangle 3: (3 × 2) + 3 = 9. Hence, the correct number is 9.

[CampusMonk Module 11 (Number Puzzles Q2)]
Directions: *Concentric Segmented Circle Sums.*



Q19. Which number replaces the question mark in the upper right circle?

a. 1
b. 2
c. 3
d. 4

Answer: c. 3

Logic / Trick: Using the lower two circles as a source, the values in corresponding segments of the upper left circle equal the sums of the numbers in the lower two circles. Matching corresponding segments yields ? = 3.

[CampusMonk Module 11 (Number Puzzles Q3)]
Directions: *Bi-directional Horizontal Sequence.*



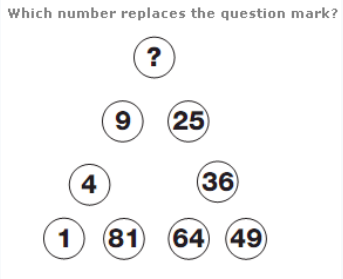
Q20. Which number replaces the question mark in the central circle?

a. 2
b. 6
c. 9
d. 12

Answer: b. 6

Logic / Trick: Starting on the extreme left and right hand of the row and working towards the centre, add the end digits together and subtract 3 to give the next value along to the left/right. Converging inwards yields 6.

[CampusMonk Module 11 (Number Puzzles Q4)]
Directions: *Triangular Square Sequence.*



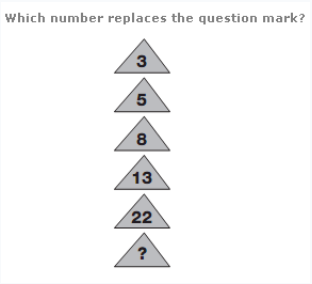
Q21. Which number replaces the question mark along the perimeter of the triangle?

a. 16
b. 20
c. 24
d. 28

Answer: a. 16

Logic / Trick: Starting bottom left and moving clockwise around the triangle, numbers follow the sequence of square numbers: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$. Hence, ? = 16.

[CampusMonk Module 11 (Number Puzzles Q5)]
Directions: *Vertical Cascading Double-and-Subtract.*



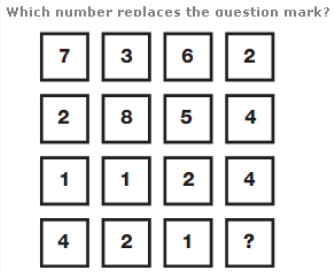
Q22. Which number replaces the question mark at the base of the tower?

a. 21
b. 25
c. 30
d. 39

Answer: d. 39

Logic / Trick: Working from top to bottom, double each number and subtract 1, then 2, then 3, then 4: 21 × 2 - 3 = 39. Hence, ? = 39.

[CampusMonk Module 11 (Number Puzzles Q6)]
Directions: *Column Constant Sum Matrix.*



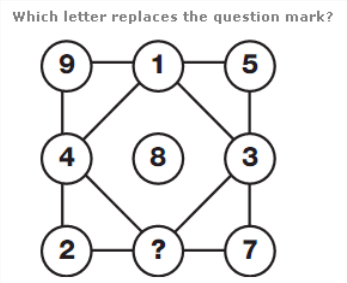
Q23. Which number replaces the question mark in the third column?

a. 1
b. 2
c. 3
d. 4

Answer: d. 4

Logic / Trick: Working in columns, the sum of the numbers in each column is always exactly 14. For the target column: 14 - sum(known) = 4. Hence, ? = 4.

[CampusMonk Module 11 (Number Puzzles Q7)]
Directions: *3×3 Magic Grid Sum.*



Q24. Which number replaces the question mark in the grid?

a. 6
b. 12
c. 18
d. 24

Answer: a. 6

Logic / Trick: The numbers in each row and column add up to 15. The missing cell requires 15 - (known row sum) = 6. Hence, ? = 6.

[CampusMonk Module 11 (Number Puzzles Q8)]
Directions: *Horizontal Incremental Difference.*



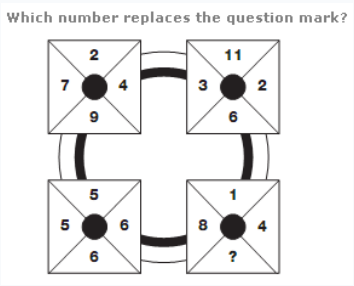
Q25. Which number replaces the question mark in the circle sequence?

a. 35
b. 40
c. 45
d. 50

Answer: b. 40

Logic / Trick: Moving from left to right, consecutive numbers increase by +2, +3, +4, and +5: 35 + 5 = 40. Hence, ? = 40.

[CampusMonk Module 11 (Number Puzzles Q9)]
Directions: *Symmetrical Sub-Square Sums.*



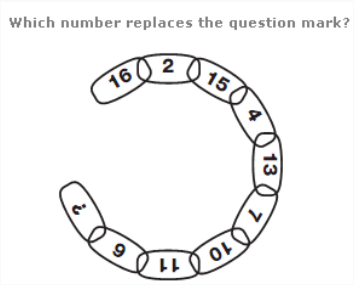
Q26. Which number replaces the question mark in the bottom-right quadrant?

a. 3
b. 6
c. 9
d. 12

Answer: c. 9

Logic / Trick: In each square of the diagram, the sum of the numbers is always constant and equals 22. In the target square: 22 - (sum of three entries) = 9. Hence, ? = 9.

[CampusMonk Module 11 (Number Puzzles Q10)]
Directions: *Alternating Interleaved Segment Chain.*



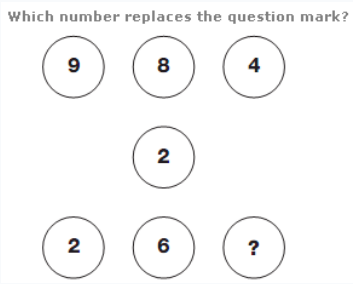
Q27. Which number replaces the question mark in the chain?

a. 16
b. 22
c. 26
d. 29

Answer: a. 16

Logic / Trick: Moving clockwise around alternate segments, one sequence decreases by -1, -2, -3, -4 while the other increases by +2, +3, +4, +5. Evaluating the target step yields 16.

[CampusMonk Module 11 (Number Puzzles Q11)]
Directions: *Central Linear Intersection Differences.*



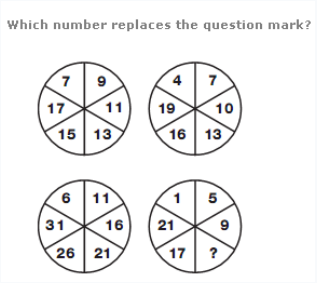
Q28. Which number replaces the question mark in the lower segment?

a. 4
b. 5
c. 6
d. 7

Answer: d. 7

Logic / Trick: Following straight diameter lines through the centre of the diagram, subtract the middle number from the top number to yield the corresponding bottom number: Top - Centre = Bottom. Target value = 7.

[CampusMonk Module 11 (Number Puzzles Q12)]
Directions: *4-Quadrant Circular Clocks.*



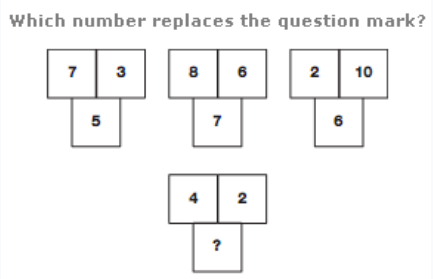
Q29. Which number replaces the question mark in the lower quadrant?

a. 10
b. 13
c. 16
d. 19

Answer: b. 13

Logic / Trick: In each circle, starting at the top-left segment, numbers increase moving clockwise: Upper-left (+2), Upper-right (+3), Lower-right (+4), and Lower-left (+5). The missing value is 13.

[CampusMonk Module 11 (Number Puzzles Q13)]
Directions: *Triad Arithmetic Mean.*



Q30. Which number replaces the question mark at the apex of the triad?

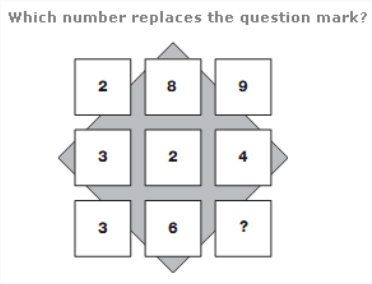
a. 1
b. 2
c. 3
d. 4

Answer: c. 3

Logic / Trick: In each group of 3 numbers, the lower number equals the arithmetic average of the top two numbers: Lower = (Top1 + Top2) / 2. Target evaluates to 3.

[CampusMonk Module 11 (Number Puzzles Q14)]

Directions: *3-Digit Square Number Grid.*



Q31. Which digit replaces the question mark in the middle row?

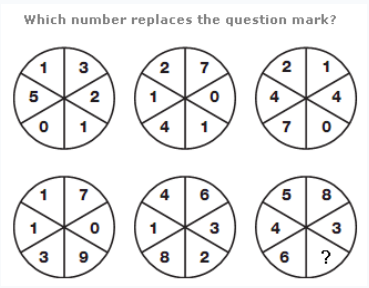
a. 1
b. 3
c. 4
d. 6

Answer: a. 1

Logic / Trick: Reading each row horizontally as a 3-digit number, the rows form consecutive squares: $17^2 = 289$, $18^2 = 324$, $19^2 = 361$. The target cell corresponds to the leading digit 1. Hence, ? = 1.

[CampusMonk Module 11 (Number Puzzles Q15)]

Directions: *Progressive Circle Sum Series.*



Q32. Which number replaces the question mark in the final circle?

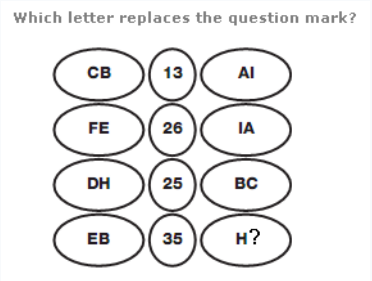
a. 4
b. 3
c. 2
d. 1

Answer: d. 1

Logic / Trick: Starting with the top-left circle and moving right across rows, the sum of numbers in each circle forms the arithmetic progression 12, 15, 18, 21, 24. Setting the circle sum to 24 yields ? = 1.

[CampusMonk Mock Day 06 (Logical Puzzle Q1)]

Directions: *Triple Oval Number-Letter Subtraction.*



Q33. Which letter replaces the question mark in the right oval?

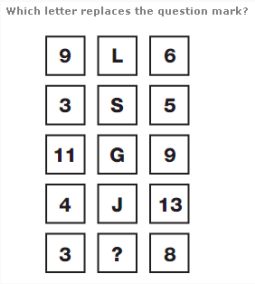
a. I
b. G
c. B
d. T

Answer: b. G

Logic / Trick: Two-digit numerical values formed by letters in outer ovals relate to the centre number by subtraction: Row 1: CB (32) - AI (19) = 13. Row 2: IA (91) - FE (65) = 26. Row 3: DH (48) - BC (23) = 25. Row 4: Right oval - EB (52) = 35 => Right oval = 52 + 35 = 87. Since H = 8, the second letter is position 7 = G.

[CampusMonk Mock Day 06 (Logical Puzzle Q2)]

Directions: *Alphabet Position Constant Row Sum 27.*



Q34. Which letter replaces the question mark in the central column?

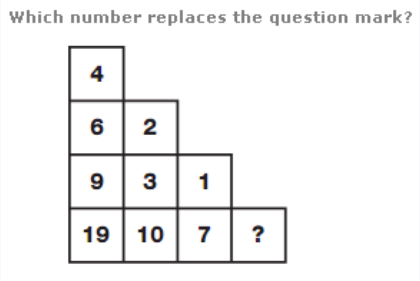
a. P
b. S
c. R
d. Q

Answer: a. P

Logic / Trick: In each row, Left Number + Alphabet Positional Value + Right Number = 27: Row 1: 9 + 12 (L) + 6 = 27. Row 2: 3 + 19 (S) + 5 = 27. Row 3: 11 + 7 (G) + 9 = 27. Row 4: 4 + 10 (J) + 13 = 27. Row 5: 3 + ? + 8 = 27 => ? = 27 - 11 = 16 = P. Correct option: P.

[CampusMonk Mock Day 06 (Logical Puzzle Q4)]

Directions: *Staircase Cascading Difference Grid.*



Q35. Which number replaces the question mark in the bottom row?

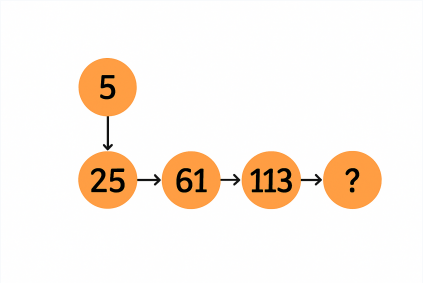
a. 5
b. 6
c. 8
d. 2

Answer: b. 6

Logic / Trick: In the bottom row, the horizontal difference between adjacent numbers equals the cell directly above in the row above: 19 - 10 = 9; 10 - 7 = 3; 7 - ? = 1 => ? = 7 - 1 = 6. Correct option: 6.

[CampusMonk Mock Day 07 (Puzzle Solving Q1)]

Directions: *Second-Order Constant Difference Node Sequence.*



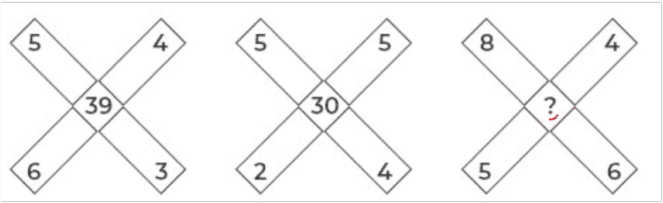
Q36. Find the value of the question mark (?) in the sequence: 5 -> 25 -> 61 -> 113 -> ?

a. 169
b. 181
c. 197
d. 205

Answer: b. 181

Logic / Trick: First differences: 25 - 5 = 20; 61 - 25 = 36 (+16); 113 - 61 = 52 (+16). The second difference is constant at +16. Next first difference = 52 + 16 = 68. Next term = 113 + 68 = 181. Correct option: 181.

[CampusMonk Mock Day 07 (Puzzle Solving Q3)]
Directions: *Diagonal Cross Wings Product Sum.*



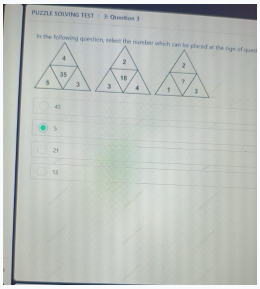
Q37. Select the number which can be placed at the question mark in the center of the third X figure.

a. 76
b. 58
c. 68
d. 98

Answer: c. 68

Logic / Trick: In each X figure, the centre number equals the sum of the products of opposite diagonal wings: Centre = (Top-Left × Bottom-Right) + (Top-Right × Bottom-Left). Figure 1: (5 × 3) + (4 × 6) = 15 + 24 = 39. Figure 2: (5 × 4) + (5 × 2) = 20 + 10 = 30. Figure 3: (8 × 6) + (4 × 5) = 48 + 20 = 68. Correct option: 68.

[OnlineStudy4U Slide Test 3 (Triangle Pattern)]
Directions: *Triangle Lateral Multiplication.*



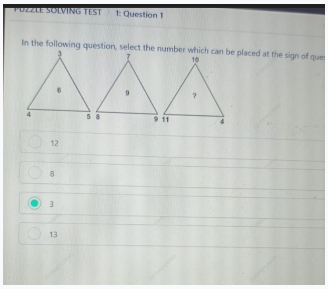
Q38. In the following question, select the number which can be placed at the sign of question mark (?) in Triangle 3.

a. 43
b. 5
c. 21
d. 13

Answer: b. 5

Logic / Trick: In each triangle: (Top + Right) × Left = Centre. Triangle 1: (4 + 3) × 5 = 7 × 5 = 35. Triangle 2: (2 + 4) × 3 = 6 × 3 = 18. Triangle 3: (2 + 3) × 1 = 5 × 1 = 5. Correct option: 5.

[OnlineStudy4U Slide Test 1 (Triangle Difference)]
Directions: *Triangle Difference Formula.*



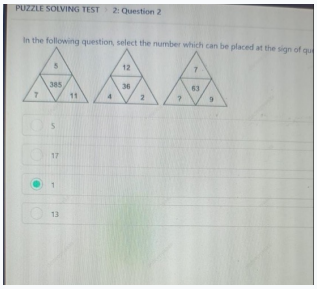
Q39. Select the number which can be placed at the question mark in Triangle 3.

a. 12
b. 8
c. 3
d. 13

Answer: c. 3

Logic / Trick: In each triangle: (Top + Right) - Left = Centre. Triangle 3: (10 + 4) - 11 = 14 - 11 = 3. Correct option: 3.

[OnlineStudy4U Slide Test 2 (Triangle Product)]
Directions: *Triangle Product Rule.*



Q40. Select the number which can be placed at the sign of question mark (?) in Triangle 3.

a. 5
b. 17
c. 1
d. 13

Answer: c. 1

Logic / Trick: In each triangle: Top × Left × Right = Centre. Triangle 1: 5 × 7 × 11 = 385. Triangle 3: 7 × ? × 9 = 63 => ? = 63 / 63 = 1. Correct option: 1.

[OnlineStudy4U Slide Pattern (4 Blocks of Squares)]
Directions: *4 Blocks of Squares Sum and Progression.*

The numbers in the following 4 blocks of squares have been filled according to some rule. Fill the number in the square with (?) mark.

1	6	4	8	7	11	10	15
4	5	6	7	8	10	10	?

(1) 13
(2) 14
(3) 16
(4) 17

Q41. The numbers in the 4 blocks of squares are filled according to a rule. Fill the number in the square with (?) mark. Block 1: [1, 6; 4, 5] | Block 2: [4, 8; 6, 7] | Block 3: [7, 11; 8, 10] | Block 4: [10, 15; 10, ?]

a. 13
b. 14
c. 16
d. 17

Answer: b. 14

Logic / Trick: Method 1 (Block Sums): Sum of Block 1 = 16 = 4^2; Sum of Block 2 = 25 = 5^2; Sum of Block 3 = 36 = 6^2; Sum of Block 4 = 49 = 7^2 => 10 + 15 + 10 + ? = 49 => ? = 14. Method 2 (Sequence): Bottom-Right cells advance by +2, +3, +4: 5 (+2) -> 7 (+3) -> 10 (+4) -> 14. Correct option: 14.

[OnlineStudy4U Slide Matrix (4×4 Division Logic)]
Directions: *4×4 Matrix Division Logic.*

54. In the following question, a figure carrying certain numbers is given. Assuming that the numbers in the figure follow a similar pattern, find the missing number.

17	8	5	5
13	7	5	4
6	12	6	3
10	6	4	?

(1) 4
(2) 5
(3) 6
(4) 2

Q42. Assuming that the numbers in the figure follow a similar pattern, find the missing number ? Row 1: [17, 8, 5, 5] Row 2: [13, 7, 5, 4] Row 3: [6, 12, 6, 3] Row 4: [10, 6, 4, ?]

a. 4
b. 5
c. 6
d. 2

Answer: a. 4

Logic / Trick: Row rule: (Col 1 + Col 2) / Col 3 = Col 4. Row 1: (17 + 8) / 5 = 25 / 5 = 5. Row 2: (13 + 7) / 5 = 20 / 5 = 4. Row 3: (6 + 12) / 6 = 18 / 6 = 3. Row 4: (10 + 6) / 4 = 16 / 4 = 4. Correct option: 4.